# AIKDAP

*AI Knowledge Discovery and Analytics Platform*
*A multi-agent "AI Work Operating System" for completing research and business analytics workflows from start to finish*
*Thesis in Brief*

## 1. Problem Statement

Researchers, students and analysts split one piece of work across many tools. They search in Google or Perplexity, read in NotebookLM, reason in ChatGPT, analyse in Kaggle and Python, and present in Power BI. Each switch loses context, nothing is remembered between sessions, and the outputs are scattered and hard to verify. Today's AI assistants answer questions, but they don't finish the work.

## 2. Thesis Statement

One workspace organised around projects and run by a multi-agent system can take a single task from the user, then plan it, split it into steps, run them, check the results and deliver the finished work. Every step stays traceable and explainable, and every output is kept as a reusable asset.

## 3. Objectives

1. Build one workspace that replaces the fragmented set of research and analytics tools.
2. Break user tasks into steps automatically using an Intent Analysis Agent and a Planner Agent.
3. Coordinate specialised agents through LangGraph, with all communication going through shared state.
4. Keep memory per project rather than per chat: Redis for short-term memory, and PostgreSQL plus ChromaDB for long-term memory.
5. Store every generated artifact as a lasting asset: reports, datasets, charts and summaries.
6. Make the system trustworthy through a Validation Service and an Explainable AI trace.
7. Adapt the output to each user persona (Student, Researcher, Project Builder).

## 4. System Architecture

```
User → Authentication → Intent Analysis Agent → Planner Agent
     → LangGraph Orchestrator → Specialized Agents → Tool Registry
     → Knowledge Layer → Response Aggregator → Validation Service
     → Explainable AI → Final Response
```

Every project contains an Overview, Research Workspace, Business Analytics, Knowledge Base, Assets, Reports, Workflow Timeline and Settings. The **Command Center** homepage answers

*"What is happening?"* rather than *"What do you want to ask?"*

## 5. Technology Stack

- **Backend:** Python 3.12, FastAPI, SQLAlchemy 2, Alembic, PostgreSQL, Redis, Celery, Pydantic v2
- **AI:** LangGraph, LangChain, OpenAI-compatible LLMs, ChromaDB (vector store)
- **Frontend:** React, TypeScript, Vite, Tailwind CSS, shadcn/ui, TanStack Query
- **Infrastructure:** Docker and Docker Compose

## 6. Key Modules

- **Research Workspace:** finds literature, analyses evidence across papers, detects contradictions and suggests papers through OpenAlex.
- **Business Analytics:** imports datasets (including from Kaggle), runs automated analysis and generates charts.
- **Knowledge Base and Assets:** stores documents, retrieves them through embeddings and keeps artifacts permanently.
- **Explainable AI:** records each run's planning, agent execution, tool usage, validation, aggregation and timing.
- **Persona Features:** produces Study Summaries, Project Synopses (DOCX/PDF) and Build Plans.

## 7. Methodology

The project was built incrementally in 11 milestones (0–10). It started with the backend foundation, then added authentication, projects, memory, AI orchestration, the research and analytics modules, explainability, the frontend, deployment and persona features. Each milestone had to be production-ready and independently testable before the next one began.

## 8. Contributions

- A planner-driven multi-agent architecture in which agents never call each other directly.
- Memory scoped to projects, which gives continuity across sessions.
- Validation and explainability treated as core parts of the pipeline rather than optional add-ons.
- A single platform that combines research and business analytics.

## 9. Conclusion

AIKDAP moves AI from *answering* to *completing*. Planning, coordinated agents, validation and lasting project memory together reduce tool fragmentation and produce work that is verifiable and reusable.